InHand Networks



# InHand 5G & 4G Routers

Secure, Reliable IoT Connectivity




## About Us

InHand Networks is a leading IoT solutions provider founded in 2001, dedicated to driving digital transformation across industries and empowering customers to unlock their full potential and achieve accelerated growth.

We specialize in delivering industrial-grade connectivity solutions for diverse sectors, such as enterprise networks, industrial and building IoT, digital energy, smart commerce, and mobility. Our comprehensive product portfolio and services cater to various applications worldwide, including smart manufacturing, smart grid, intelligent transportation, smart retail, etc. With a global footprint spanning over 60 countries, we serve customers in China, the United States, France, Germany, the United Kingdom, Italy, and beyond.

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# Contents


**Business Routers**

FWA12 5G 02
CPE02 5G 04
ER815 5G 06
ER615 08
Selection Guide 10

**Outdoor Routers**

ODU12 5G 14
ODU302 16
Selection Guide 18

**Battery-Powered Routers**

CR602 5G 22
CR202-Lite 24
Selection Guide 26

**Vehicle Routers**

VR624 5G 30

**Industrial Routers**

IR925 5G 34
IR624 5G 36
IR602 REDCap 38
IR302 40
IR315 42
Selection Guide 44

**Cloud Service**

InCloud Manager 48
DeviceLive 50
Device Manager 52
InConnect 54

# Business Routers



**FWA12 / CPE02 / ER615 / ER815**

Indoor connectivity for branches, stores, offices, and temporary business sites. The range spans high-capacity 5G CPE, Wi-Fi 7 access, multi-port enterprise routing, SD-WAN, and centralized cloud management.



# FWA12

## Indoor 5G Wi-Fi 7 Router

### 5G Release 16 performance

Supports 5G SA/NSA with theoretical peak rates up to 7.01 Gbps downlink and 2.5 Gbps uplink.

### Wi-Fi 7 for dense indoor networks

BE5000 dual-band Wi-Fi supports up to 128 wireless clients for high-capacity business access.

### One-click POS traffic priority

POS Ready prioritizes transaction traffic to help protect payment workflows when other applications compete for bandwidth.

### Centralized cloud operations

InCloud Manager supports zero-touch provisioning, visual monitoring, remote maintenance, and AI-assisted device operations.

FWA12 combines 5G Release 16, Wi-Fi 7, multi-gigabit Ethernet, and cloud management in one indoor platform. It is designed for stores, offices, and service locations that need high throughput, broad client capacity, and centrally managed operations.

## Hardware

| | |
|---|---|
| Stateful Firewall Throughput | 2 Gbps |
| Recommended Users | 220 (128 Wi-Fi devices) |
| Ethernet | 2 × 2.5 GbE RJ45, WAN/LAN switchable, dual-LAN |
| USB | 1 × Type-C 2.0, HOST and SLAVE modes |
| SIM | 2 x Nano SIM; 1 x eSIM optional |
| Reset | Reset button |
| Power Switch | 1 × button |
| LED | System, Cellular, Signal, WAN, LAN, Wi-Fi |
| Antenna | 6 × external + 2 × built-in antenna;<br>3 × built-in Wi-Fi antenna |
| Data Rate | 5G SA/NSA: 7.01 Gbps DL / 2.5 Gbps UL;<br>4G CAT19: 1.6 Gbps DL / 200 Mbps UL |
| Wi-Fi Frequency | 2.4 GHz, 5.8 GHz |
| Wi-Fi Max Bandwidth | 5000 Mbps |
| Wi-Fi Protocol | 802.11 be/ax/ac/b/g/n, Wi-Fi 7 |

| | |
|---|---|
| Input | Circular interface, 12 V / 3 A |
| Power Consumption | ≤ 24 W |
| Dimensions | 236 × 172.5 × 58.5 mm (9.29 × 6.79 × 2.3 in) |
| Weight | 2.2 lbs (1.0 kg) |
| Installation | Wall mount, Desktop mount |
| Operating Temperature | -10 °C ~ +50 °C (14 °F ~ +122 °F) |
| Storage Temperature | -40 °C ~ +85 °C (-40 °F ~ +185 °F) |
| Humidity | 5–95 % RH (non-condensing) |
| Protection | IP20 |
| Shock | IEC 60068-2-27 |
| Vibration | IEC 60068-2-6 |
| Free Fall | IEC 60068-2-32 |
| Certification | FCC, IC, PTCRB, Verizon, T-Mobile, AT&T, CE* |
| Warranty | 3 years |

## Software

| | |
|---|---|
| Platform | InCloud Manager |
| Cloud Management | Zero-touch provisioning, centralized management, visualized monitoring |
| AI Assistant | Supported through InCloud Manager |
| WAN Access | 5G/4G cellular, Ethernet |
| Link Backup | Packet-by-packet load balancing, link priority adjustment |
| Dialing | PPPoE, cellular auto-redial, dual-SIM switching, APN configuration |
| Link Monitoring | Real-time delay, jitter, packet loss, and throughput monitoring |
| IP & Network Services | IPv4/IPv6, VLAN, DHCP Server/Client, DNS, DDNS, fixed address, IP passthrough |
| VPN | IPsec VPN, L2TP VPN |
| Firewall | Access control, port mapping, port forwarding, MAC/IP/port/protocol filtering |
| Dashboard | Device info, interface status, traffic analysis |
| Logs | System logs, diagnostic logs, device events, email alerts |
| Cellular Monitoring | Cellular signal monitoring |
| Mode | 802.11 be/ax/ac/b/g/n, Wi-Fi 7, AP mode |
| Policy | Policy routing, traffic shaping |
| POS Ready™ | One-click POS traffic optimization; prioritizes bandwidth for mission-critical transactions |
| Watchdog | Software and hardware watchdog, self-healing |
| Management Access | Web UI, CLI remote access and control |
| Remote Maintenance | Remote maintenance of PCs, monitoring equipment, servers, etc. |
| Cloud Tools | Remote diagnostics and configuration |



# CPE02

## Indoor 5G Wi-Fi 6 Router

### Fast 5G access

Supports 5G SA/NSA with downlink rates up to 3.4Gbps for bandwidth-intensive indoor applications.

### Wi-Fi 6 AX5400

Dual-band Wi-Fi 6 supports up to 128 wireless clients, multiple SSIDs, VLANs, and WPA2 security.

### Cellular and wired failover

Link detection and WAN backup help maintain service when the preferred uplink becomes unavailable.

### Simplified fleet management

InCloud Manager provides remote onboarding, configuration, upgrades, monitoring, and maintenance for distributed sites.

CPE02 converts 5G or 4G connectivity into high-capacity Wi-Fi 6 access for retail, office, and shared indoor environments. Regional Ethernet options and centralized cloud management support flexible deployment across business networks.

## Hardware

| | |
|---|---|
| CPU | IPQ5018 |
| Firewall Throughput | 1 Gbps |
| Recommended Users | 200 (Wi-Fi: 128) |
| Cellular(CPE02-NANR) | 5G NSA: 3.4 Gbps DL / 550 Mbps UL;<br>5G SA: 2.4 Gbps DL / 900 Mbps UL;<br>4G LTE: 1.6 Gbps DL / 200 Mbps UL |
| Cellular(CPE02-EUNR) | 5G NSA: 2.6 Gbps DL / 650 Mbps UL;<br>5G SA: 2.0 Gbps DL / 1.0 Gbps UL;<br>4G LTE: 600 Mbps DL / 150 Mbps UL |
| Ethernet | 2 × GbE (CPE02-EUNR) or<br>2.5G + 1G (CPE02-NANR),<br>WAN/LAN and dual LAN support |
| SIM Card | 1 × Nano 4FF, 1 × eSIM optional |
| Type-C | Debug only |
| Reset | Hardware factory reset |
| WPS | Client one-touch Wi-Fi connection |
| Standard | Wi-Fi 6, 802.11 a/b/g/n/ac/ax |
| Wi-Fi Max Rate(CPE02-NANR) | 5400 Mbps |
| Wi-Fi Max Rate(CPE02-EUNR) | 3000 Mbps |
| TX Power | 2.4 GHz: 17 dBm; 5 GHz: 17 dBm |
| Antenna Gain | ≤ 5 dBi |
| Input | 12 V / 2 A |
| Power Consumption | ≤ 15 W |
| LED | 5G, Wi-Fi, Signal, Power |
| Dimensions (L × W × H) | 105 × 105 × 217 mm (4.13 × 4.13 × 8.54 in) |
| Weight | 800 g (1.76 lb) |
| Installation | Desktop, wall-mount |
| Protection | IP20 |
| Operating Temperature | -10 °C ~ +40 °C (14 °F ~ 104 °F) |
| Storage Temperature | -40 °C ~ +85 °C (-40 °F ~ 185 °F) |
| Humidity | 5 % ~ 95 % RH (non-condensing) |
| Certification | CE, FCC, IC, PTCRB, AT&T, Verizon,<br>T-Mobile |
| EMC | EMC Level 2 |
| Warrant | 3 years |

## Software

| | |
|---|---|
| Platform | InCloud Manager |
| Cloud Management | Unified device access, zero-touch remote deployment, bulk remote upgrades,<br>configuration deployment, connector remote maintenance, two-factor authentication |
| Dashboard | Supported |
| WAN Access | 5G/4G, wired |
| Intelligent Links | Real-time link detection |
| Network Functions | Dialing; IP Protocols; Network Protocols; Speed Test; VPN; Routing |
| Wi-Fi Features | Multiple SSID, SSID VLAN, SSID hidden |
| Encryption | WPA, WPA2, WPA3* |
| Upgrade | Scheduled upgrade |
| Logs | Runtime logs, diagnostic logs |
| Monitoring and Diagnostics | Events; Alerts; Diagnostics |




# ER815

## 5G Enterprise Router with 2.5GbE and Wi-Fi 6

### High-speed 5G access

Supports 5G SA/NSA with theoretical downlink rates up to 3.4 Gbps, depending on operating mode and network.

### 2.5GbE uplink capacity

A 2.5GbE WAN option adds wired bandwidth headroom for high-speed broadband and multi-WAN deployments.

### Wi-Fi 6 for business users

AX3000 dual-band Wi-Fi supports multiple SSIDs and an estimated 150 to 200 users across the network.

### Secure branch interconnection

SD-WAN and a broad VPN set support controlled connectivity between branches, remote users, and central services.

ER815 is designed for business sites that need more WAN headroom and stronger encrypted-tunnel performance. It combines 5G, 2.5GbE, Wi-Fi 6, multi-WAN flexibility, SD-WAN, and centralized management.

## Hardware

| | |
|---|---|
| Throughput | 1 Gbps |
| IPsec VPN Throughput | 300 Mbps |
| Recommended Users | 150–200 |
| RAM | 512 MB |
| Flash | 8 GB |
| Cellular | 5G SA: 2.4 Gbps DL / 900 Mbps UL;<br>5G NSA: 3.4 Gbps DL / 550 Mbps UL;<br>4G LTE: 1.6 Gbps DL / 200 Mbps UL |
| Ethernet | WAN: 1 × 2.5G or 1 × 2.5G + 1 × GbE;<br>LAN: 4 × GbE or 3 × GbE;<br>WAN/LAN switching, dual WAN |
| SIM | 2 x Nano SIM; 1 x eSIM optional |
| Reset | Pinhole reset button |
| Antenna | 4G: SMA × 2, Wi-Fi: RP-SMA × 2;<br>5G: SMA × 4, Wi-Fi: RP-SMA × 2 |
| Standard | 802.11 a/b/g/n/ac/ax |
| Wi-Fi Max Rate | 3000 Mbps |
| Wi-Fi TX Power | 2.4 GHz: 21 dBm; 5 GHz: 21 dBm |
| Antenna Gain | ≤ 5 dBi |
| Input | 12 V / 3 A |
| Peak Power | ≤ 18 W |
| LED | Power, Network, 5G Wi-Fi, 2.4G Wi-Fi |
| Dimensions | 210 × 140 × 35 mm (8.27 × 5.51 × 1.38 in) |
| Weight | 1.25 kg (2.76 lb) |
| Installation | Bracket mount, wall mount |
| Protection | IP20 |
| Operating Temperature | -10 °C ~ +50 °C (14 °F ~ +122 °F) |
| Storage Temperature | -40 °C ~ +85 °C (-40 °F ~ +185 °F) |
| Humidity | 5–95 % RH (non-condensing) |
| Certification | CE, FCC, IC, PTCRB, AT&T,<br>Verizon, T-Mobile |
| EMC | EMC level 2 |
| Warrant | 3 years |

## Software

| | |
|---|---|
| Platform | InCloud Manager |
| Cloud Management | Unified device access, zero-touch remote deployment, bulk remote upgrades, configuration deployment,<br>SD-WAN networking, Connector remote maintenance, two-factor authentication |
| Dashboard | Supported |
| WAN Access | 5G/4G, wired, Wi-Fi |
| Intelligent Links | Real-time link detection |
| Network Functions | Dialing; IP Protocols; Protocols; VPN; SD-WAN; Routing |
| Wi-Fi Features | Multi-SSID, SSID VLAN, SSID hidden, guest mode, custom splash portal |
| Encryption | WPA, WPA2, WPA-PSK, WPA2-PSK |
| Firewall | 3L inbound/outbound rules, port forwarding, SNAT, DNAT |
| Remote access control | Supported |
| Access Control | Supported |
| Traffic Shaping | QoS by link, IP, and protocol |
| Upgrades | Scheduled upgrades |
| Monitoring and Alerts | Logs; Events; Alarms |
| Tools | ICMP, packet capture, tracert |




# ER615

## 4G LTE Wi-Fi 6 Enterprise Router

### Flexible 5G and wired access

Combines cellular and wired broadband with link backup and load balancing for resilient branch connectivity.

### Wi-Fi 6 AX3000

Dual-band Wi-Fi 6 supports up to 128 wireless clients, multiple SSIDs, VLANs, and guest access.

### Five Gigabit Ethernet ports

Five configurable Gigabit ports support dual-WAN designs and wired access for business terminals.

### Cloud-managed SD-WAN

InCloud Manager supports SD-WAN branch connectivity, zero-touch deployment, monitoring, and remote maintenance.

ER615 combines 4G LTE, Wi-Fi 6, five Gigabit Ethernet ports, and SD-WAN in a compact router for flexible branch networking and centralized management.

## Hardware

| | |
|---|---|
| Throughput | 1 Gbps |
| IPsec VPN Throughput | 100 Mbps |
| Recommended Users | 200 (Wi-Fi: 128) |
| RAM | 512 MB |
| Flash | 8 GB |
| Cellular | 4G LTE: 150 Mbps DL / 50 Mbps UL |
| Ethernet | 5 × 10/100/1000 Mbps, WAN/LAN switching, dual WAN |
| SIM | 2 x Nano SIM; 1 x eSIM optional |
| Reset | Pinhole reset button |
| Antenna | 4G: SMA × 2; Wi-Fi: RP-SMA × 2 |
| Standard | 802.11 a/b/g/n/ac/ax |
| Wi-Fi Max Rate | 3000 Mbps |
| TX Power | 2.4 GHz: 18 dBm; 5 GHz: 17 dBm |
| Antenna Gain | ≤ 5 dBi |

| | |
|---|---|
| Input | 12 V / 2 A |
| Peak Power | ≤ 24 W |
| LED | Power, Network, Signal, Wi-Fi |
| Dimensions | 198 × 137 × 25 mm (7.8 × 5.39 × 0.98 in) |
| Weight | 848 g (1.87 lb) |
| Installation | Desktop, wall-mount |
| Protection | IP20 |
| Operating Temperature | -10 °C ~ +50 °C (14 °F ~ +122 °F) |
| Storage Temperature | -40 °C ~ +85 °C (-40 °F ~ +185 °F) |
| Humidity | 5–95 % RH (non-condensing) |
| Certification | CE |
| EMC | EMC level 2 |
| Warrant | 3 years |

## Software

| | |
|---|---|
| Platform | InCloud Manager |
| Cloud Management | Centralized configuration, zero-touch deployment, remote upgrades, SD-WAN |
| Dashboard | Supported |
| WAN Access | 5G/4G, wired, Wi-Fi |
| IP Protocols | IPv4, IPv6 |
| Network Functions | Dialing; Protocols; VPN; SD-WAN; Routing |
| Wi-Fi Features | Multi-SSID, SSID VLAN, SSID hidden, guest mode |
| Encryption | WPA, WPA2, WPA-PSK, WPA2-PSK, WPA3 |
| Firewall | 3L inbound/outbound rules, port forwarding, SNAT, DNAT |
| Access Control | Black/white list, domain filtering |
| Traffic Shaping | Traffic shaping by link/IP/protocol |
| Upgrades | Scheduled upgrades |
| Monitoring and Alerts | Logs; Events; Alarms |
| Tools | ICMP, packet capture, traceroute |

# Selection Guide

| | Model | FWA12 | CPE02 | ER815 | ER615 |
|---|---|---|---|---|---|
| Performance | Stateful firewall throughput | 2 Gbps | 1 Gbps | 1 Gbps | 1 Gbps |
| | Recommended users | 220 total / 128 Wi-Fi | 200 total / 128 Wi-Fi | 150–200 total | 200 total / 128 Wi-Fi |
| Interface | Cellular | 5G SA/NSA: up to 7.01 Gbps DL; LTE Cat.19 | 5G SA/NSA: up to 3.4 Gbps DL; 4G LTE | 5G SA/NSA: up to 3.4 Gbps DL; LTE | 4G LTE: up to 150 Mbps DL / 50 Mbps UL |
| | Ethernet | 2 × 2.5GbE, WAN/LAN switchable | 2.5GbE + GbE | 1 x 2.5GbE+4*GBE, WAN/LAN switchable | 5 × GbE, WAN/LAN switchable |
| | Wi-Fi standard | Wi-Fi 7 (802.11be) | Wi-Fi 6 (802.11ax) | Wi-Fi 6 (802.11ax) | Wi-Fi 6 (802.11ax) |
| | Maximum Wi-Fi rate | 5,000 Mbps | 5,400 Mbps | 3,000 Mbps | 3,000 Mbps |
| | SIM | 2 x Nano SIM + 1 x eSIM optional | 1 x Nano SIM + 1 x eSIM optional | 2 x Nano SIM + 1 x eSIM optional | 2 x Nano SIM + 1 x eSIM optional |
| | Antenna configuration | 6 × External + 2 × Built-in Cellular; 3 × built-in Wi-Fi | 4 × Built-in Cellular + 3 × Built-in Wi-Fi | 5G: 4 × SMA; Wi-Fi: 2 × RP-SMA | 4G: 2 × SMA; Wi-Fi: 2 × RP-SMA |
| Power | Power input | 12 V / 3 A | 12 V / 2 A | 12 V / 3 A | 12 V / 2 A |
| | Maximum power consumption | ≤ 24 W | ≤ 15 W | ≤ 18 W | ≤ 24 W |
| Mechanical | Dimensions | 236 × 172.5 × 58.5 mm | 105 × 105 × 217 mm | 210 × 140 × 35 mm | 198 × 137 × 25 mm |
| | Weight | 1kg | 800 g | 1.25 kg | 848 g |
| | Mounting | Desktop / wall | Desktop / wall | Bracket / wall | Desktop / wall |
| | Protection rating | IP20 | IP20 | IP20 | IP20 |
| Environment | Operating temperature | -10 °C to +50 °C | -10 °C to +40 °C | -10 °C to +50 °C | -10 °C to +50 °C |
| | Storage temperature | -40 °C to +85 °C | -40 °C to +85 °C | -40 °C to +85 °C | -40 °C to +85 °C |
| | Operating humidity | 5–95% RH, non-condensing | 5–95% RH, non-condensing | 5–95% RH, non-condensing | 5–95% RH, non-condensing |
| Software | Cloud management | InCloud Manager | InCloud Manager | InCloud Manager | InCloud Manager |
| | Link / SD-WAN functions | Link monitoring; packet-by-packet load balancing; link priority | Real-time link detection | SD-WAN; real-time link detection | SD-WAN spoke; centralized configuration |
| | VPN / security | IPsec, L2TP; firewall and traffic controls | IPsec, L2TP; OpenVPN*; WPA3* | IPsec, L2TP, VXLAN, GRE; OpenVPN*; firewall | IPsec, L2TP; firewall and access controls |
| Certification | Certification | FCC, IC, PTCRB, Verizon, T-Mobile, AT&T | CE, FCC, IC, PTCRB, AT&T, Verizon, T-Mobile | CE, FCC, IC, PTCRB, AT&T, Verizon, T-Mobile | CE |

# Outdoor Routers



## ODU12 / ODU302

All-in-one outdoor routers that combine cellular access, integrated antennas, flexible mounting, and IP67-rated protection. Choose ODU12 for high-capacity 5G and Wi-Fi 7, or ODU302 for compact 4G deployments.



# ODU12

## Outdoor 5G Wi-Fi 7 Router

### 5G Release 16

Theoretical peak rates reach 7.01 Gbps downlink and 2.5 Gbps uplink.

### Outdoor Wi-Fi 7

BE5000 dual-band Wi-Fi supports up to 128 wireless clients.

### Integrated antennas

Built-in cellular and Wi-Fi antennas reduce external components and feeder cabling.

### IP67 PoE design

PoE and wall, pole, or roof mounting simplify outdoor installation.

ODU12 integrates 5G, Wi-Fi 7, 2.5GbE, PoE, and built-in antennas in a compact IP65-rated enclosure. It is intended for outdoor locations that need strong access capacity without a separate router, antenna array, and weatherproof cabinet.

## Hardware

| | |
|---|---|
| Processor | Quad-core Cortex-A55, 2.2 GHz |
| Memory / Storage | 2 GB DDR4 / 4 GB eMMC |
| 5G Platform | MTK 830, 3GPP Release 16 |
| SIM / eSIM | 2 x Nano SIM; 1 x eSIM optional |
| Ethernet | 2 × 2.5GbE RJ45; WAN supports PoE input |
| USB | 1 × Type-C |
| Wi-Fi | Wi-Fi 7 BE5000, dual-band |
| Cellular Antennas | 8 × built-in omnidirectional antennas |
| Wi-Fi / GPS Antennas | 3 × built-in Wi-Fi; 1 × built-in GPS |
| Power | IEEE 802.3at PoE; 802.3af compatible |
| Power Consumption | ≤18 W |
| Cooling | Fanless passive cooling |
| Enclosure | Die-cast aluminum heat sink with UV-stabilized shell |

| | |
|---|---|
| Protection | IP67 |
| Operating Temperature | -30°C ~ +70°C（-22°F ~ +15°F） |
| Storage Temperature | -40°C ~ +85°C（-40°F ~ +185°F） |
| Humidity | 5–95%, non-condensing |
| Dimensions / Weight | 235 × 167.93 × 81.93 mm / approx. 1.5 kg |
| Mounting | Wall, pole, or roof |
| Certification | FCC, IC, PTCRB, CE*, AT&T, Verizon, T-Mobile |
| Environmental | RoHS, REACH |
| Salt Mist | 48 hours; IEC 60068-2-52 |
| Shock / Vibration / Free Fall | IEC 60068-2-27 / -6 / -32 |
| Warrant | 3 years |

## Software

| | |
|---|---|
| Platform | InCloud Manager |
| AI Assistant | Supported through InCloud Manager |
| Cellular | 5G SA/NSA Release 16; LTE Cat.19 |
| Peak Cellular Rates | 5G: 7.01 Gbps DL / 2.5 Gbps UL; LTE: 1.6 Gbps DL / 200 Mbps UL |
| SIM / eSIM | Dual-SIM hot backup; eSIM provisioning reserved |
| Ethernet | 2.5GbE WAN and LAN; dual-LAN mode supported |
| Wi-Fi / Security | 802.11be; WPA2/WPA3 Personal and Enterprise |
| Link Backup | Active/standby or load balancing; ICMP/TCP/DNS probes |
| Link Monitoring | Status, latency, packet loss, and anomaly alerts |
| Routing / NAT | Policy and forced routing; SNAT/DNAT with NAPT |
| Traffic Management | Queue-based shaping and critical-traffic priority |
| VPN | IPsec and L2TP |
| LAN Services | VLAN, bridge, DHCP pools, DNS proxy, IP-MAC binding |
| Portal | Local portal with username/password, SMS, or WeChat authentication |
| Management | Web interface, app configuration, multi-user permissions |
| Logs / Alerts | System logs and configurable alert rules |
| Maintenance / Location | Scheduled reboot and coarse cell-tower positioning |



# ODU302

## 4G Industrial Outdoor Router

### Integrated antennas

Built-in 4G and Wi-Fi antennas reduce accessories, cabling, and enclosure needs.

### PoE or DC power

IEEE 802.3af PoE and 9–36 VDC support flexible installation.

### Connection recovery

Dual-SIM failover, wired backup, auto-redial, and watchdog support link recovery.

### Remote outdoor operations

IP65 protection and Device Manager support distributed monitoring and maintenance.

ODU302 combines 4G LTE, integrated antennas, PoE, dual-SIM resilience, and IP65 protection in a single outdoor device. Wall, pole, and top mounting make it practical for surveillance, charging, signage, and other distributed sites.

## Hardware

| | |
|---|---|
| Processor | 580 MHz |
| Memory / Storage | 128 MB DDR2 / 32 MB SPI NOR Flash |
| Cellular | LTE Cat.4: 150 Mbps DL / 50 Mbps UL |
| Recommended Users | 32 |
| Ethernet | 2 × Fast Ethernet; WAN/LAN switching; LAN2 supports PoE |
| SIM | Dual Nano-SIM with failover |
| Digital I/O | 2 × configurable DI/DO |
| Antennas | 2 × built-in 4G; 1 × built-in Wi-Fi |
| Wi-Fi | 802.11 b/g/n, 2.4 GHz, up to 150 Mbps |
| Indicators / Reset | System, Wi-Fi, cellular, Ethernet / pin-hole reset |
| Power Input | 9–36 VDC or IEEE 802.3af PoE |

| | |
|---|---|
| Power Protection | Over-current and reverse-polarity protection |
| Power Consumption | 90–110 mA standby; 280–300 mA operating; 340–360 mA peak at 12 V |
| Enclosure / Protection | Plastic / IP65 |
| Operating Temperature | -40°C ~ +70°C（-40 °F ~ +158 °F） |
| Storage Temperature | -40°C ~ +85°C（-40 °F ~ +185 °F） |
| Humidity | 5–95%, non-condensing |
| Dimensions / Weight | 164 × 50 × 129.7 mm / approx. 500 g |
| Mounting | Wall, pole, or top |
| Shock / Vibration / Free Fall | IEC 60068-2-27 / -6 / -32 |
| Certification | FCC, IC, PTCRB, Verizon, AT&T, T-Mobile, CE |
| EMC Immunity | EN 61000-4-2/-3/-4/-5/-6/-8/-12, Level 2 |
| Warrant | 3 years |

## Software

| | |
|---|---|
| Remote Management | Status, configuration, firmware upgrade, and reboot |
| Cellular Access | 2G/3G/4G modes; APN and VPDN |
| Dual-SIM | Primary/backup switching |
| WAN / LAN | Static IP, DHCP, PPPoE, ARP, Ethernet |
| Wi-Fi / Security | 802.11 b/g/n; open/shared key, WPA/WPA2 |
| Link Backup | Cellular and wired WAN backup; ICMP/TCP/DNS detection |
| Reliability | VRRP, heartbeat monitoring, auto-reconnect, watchdog |
| Firewall / NAT | ACL and NAPT |
| VPN | IPsec, L2TP, PPTP, GRE, OpenVPN, WireGuard, ZeroTier |
| Traffic Management | Interface bandwidth limits and QoS policies |
| LAN Services | DHCP server/relay, DNS proxy, DDNS |
| Routing / Bridge | Static/default routes and IP passthrough |
| Management Access | Web, Telnet, SSH, console, SMS status query |
| Logs | Local/remote/serial logs; key logs persist through power loss |
| Maintenance | Dial-on-demand and data/SMS activation |
| SNMP | v1/v2c/v3 with Trap alarms |

# Selection Guide

| | Model | ODU12 | ODU302 |
|---|---|---|---|
| Hardware platform | Processor | Quad-core Cortex-A55 @ 2.2 GHz | 580 MHz |
| | Memory / storage | 2 GB DDR4 / 4 GB eMMC | 128 MB DDR2 SDRAM / 32 MB SPI NOR Flash |
| Interface | Cellular | 5G SA/NSA: up to 7.01 Gbps DL; LTE Cat.19 | 4G LTE Cat.4: up to 150 Mbps DL / 50 Mbps UL |
| | Ethernet | 2 × 2.5GbE | 2 × Fast Ethernet; LAN2 supports PoE input |
| | Wi-Fi | Wi-Fi 7 BE5000; up to 128 clients | 2.4 GHz Wi-Fi 4; up to 150 Mbps |
| | SIM | 2 x Nano SIM; 1 x eSIM optional | Dual Nano-SIM |
| | Digital I/O | - | 2 × configurable DI / DO |
| | Integrated antennas | 8 × cellular + 3 × Wi-Fi + GPS（GPS is only available for ODU12-NANR) | 2 × cellular + 1 × Wi-Fi |
| Power | Power input | IEEE 802.3at PoE; 802.3af compatible | IEEE 802.3af PoE or 9–36 V DC |
| | Power consumption | ≤ 18 W | 90–110 mA standby; 280–300 mA operating @ 12 V |
| Mechanical | Dimensions | 235 × 167.93 × 81.93 mm | 164 × 50 × 129.7 mm |
| | Weight | Approx. 1.5 kg | Approx. 500 g |
| | Mounting | Wall / pole / roof | Wall / pole / top |
| | Protection rating | IP67 | IP65 |
| Environment | Operating temperature | -30 °C to +70 °C | -40 °C to +70 °C |
| | Operating humidity | 5–95% RH, non-condensing | 5–95% RH, non-condensing |
| Reliability | Mechanical test standards | IEC 60068-2-6 / -2-27 / -2-32 | IEC 60068-2-6 / -2-27 / -2-32 |
| | EMC / environmental | 48-hour salt mist test; IEC 60068-2-52 | EN 61000-4-2 / -3 / -4 / -5 / -6 / -12; Level 2 where specified |
| Software | Cloud management | InCloud Manager | Device Manager |
| | Link continuity | Active/standby or load balancing; ICMP/TCP/DNS probing | Dual-SIM failover; wired/cellular primary-backup; heartbeat and watchdog |
| | VPN | IPsec, L2TP | IPsec, L2TP, PPTP, GRE, OpenVPN, WireGuard, ZeroTier |
| | Security / O&M | Firewall, policy routing, traffic shaping, logs and diagnostics | ACL, firewall, QoS, SNMP, remote configuration, logs and diagnostics |
| Certification | Certification status | FCC, IC, PTCRB, CE*, AT&T, Verizon, T-Mobile | FCC, IC, PTCRB, AT&T, Verizon, T-Mobile, CE |

# Battery-Powered Routers



### CR202-Lite / CR602

Battery-powered cellular routers for mobile teams, temporary locations, travel, and backup connectivity. The range covers compact 4G access and higher-capacity 5G with Wi-Fi 7 and extended battery runtime.




# CR602

## Battery-Powered 5G Wi-Fi 7 Router

### 5G Release 16

Theoretical peak rates reach 7.01 Gbps downlink and 2.5 Gbps uplink.

### Wi-Fi 7 BE3000

Dual-band Wi-Fi supports 32 wireless clients and up to 128 total devices.

### Extended battery runtime

The 10,000 mAh battery supports up to 12 hours in 5G mode.

### Triple-SIM resilience

Dual Nano-SIM and eSIM support prioritized backup and centralized management.

CR602 combines 5G, Wi-Fi 7, two 2.5GbE ports, triple-SIM resilience, and battery power for temporary business sites, events, emergency response, and backup access.

## Hardware

| | |
|---|---|
| Processor | Quad-core Cortex-A55, 2.2 GHz |
| Memory / Storage | 1 GB DDR4 / 4 GB eMMC |
| 5G Module | RG620T |
| SIM / eSIM | 2 x Nano SIM; 1 x eSIM optional |
| Ethernet | 2 × 2.5GbE RJ45, adaptive WAN/LAN |
| USB | 1 × USB-C for charging |
| Wi-Fi | Wi-Fi 7 BE3000, dual-band |
| Antennas | 8 × cellular (2 external, 6 built-in); 3 × built-in Wi-Fi |
| Battery | Built-in 10,000 mAh lithium battery |
| Power Supply | USB-C or battery |
| Power Consumption | ≤20 W |
| Buttons / Indicators | Power, reset / power, network, Wi-Fi, battery |
| Cooling | Passive |
| Enclosure / Protection | Engineering plastic / IP30 |
| Operating Temperature | -10°C ~ +40°C（14°F ~ +104°F） |
| Storage Temperature | -20°C ~ +60°C（-4°F ~ +140°F） |
| Humidity | 5–95%, non-condensing |
| Dimensions / Weight | 167 × 102 × 36.5 mm / approx. 550 g |
| Battery Protection | Overcharge, over-discharge, over-current, short-circuit, temperature |
| Safety Certification | UN38.3 |
| Certification | FCC, IC, PTCRB, RoHS, REACH Verizon, AT&T, T-Mobile, FirstNet, CE* |
| Warranty | 3 years (Battery: 1 year) |

## Software

| | |
|---|---|
| Access Capacity | 32 wireless clients; up to 96 wired devices via switches |
| AI Assistant | Supported through InCloud Manager |
| Cellular | 5G SA/NSA Release 16; LTE Cat.19 |
| Peak Cellular Rates | 5G: 7.01 Gbps DL / 2.5 Gbps UL |
| Wi-Fi | 802.11be, dual-band, 3×3 MIMO |
| Interfaces | 2 × 2.5GbE WAN/LAN; USB-C 2.0 |
| LAN Services | VLAN, bridge, DHCP pools, DNS proxy/filtering, IP-MAC binding |
| Routing / NAT | Static routes, DDNS, SNAT/DNAT, NAPT |
| Link Backup | Primary/backup and load balancing; ICMP detection |
| Link Monitoring | Status, latency, packet loss, and jitter |
| Traffic Management | Forced routing and queue-based shaping |
| VPN | IPsec and L2TP |
| Device Monitoring | Device, traffic, signal, and battery status |
| Management | Web interface and app-based cellular configuration |
| Logs / Alerts | System logs and configurable alert rules |
| Maintenance | Scheduled reboot and firmware upgrade |
| Location | Coarse cell-tower positioning |




# CR202-Lite

## Battery-Powered 4G Cellular Router

### Flexible 4G connectivity

A compact 235 g design with foldable antennas fits easily into mobile and temporary deployment workflows.

### Optional removable battery

The optional 3000 mAh battery supports up to eight hours of operation and can be replaced when needed.

### Wi-Fi and wired access

2.4 GHz Wi-Fi and two Fast Ethernet ports connect up to 32 users across wireless and wired devices.

### Cloud-based device management

Device Manager supports remote monitoring, configuration, firmware updates, and event alerts for distributed units.

CR202-Lite combines 4G, Wi-Fi, wired Ethernet, and optional battery power in a compact form factor. It is designed for mobile staff, pop-up operations, temporary sites, and backup access.

## Hardware

| | |
|---|---|
| Data Throughput | 100 Mbps |
| Recommended Users | 32 |
| Cellular | 4G LTE Cat.4/Cat.6; Cat.6: 300 Mbps DL / 50 Mbps UL |
| Ethernet | 2 × 10/100 Mbps, WAN/LAN switchable, dual-LAN |
| SIM | 1 × Nano-SIM, 1 × eSIM optional |
| Reset | Reset button |
| ON/OFF | ON/OFF button |
| Antenna | 2 × external 4G cellular antennas,<br>2 × internal Wi-Fi antennas |
| LED | System, Cellular, Wi-Fi, Battery |
| Frequency | Single-band 2.4 GHz |
| Max Bandwidth | 300 Mbps |
| Protocol | 802.11 b/g/n |
| Mode | AP / STA |

| | |
|---|---|
| Input | TYPE-C, 5 V / 2 A |
| Battery | 3,000 mAh optional, lasts up to 8 hours |
| Installation | Desktop, wall mounting |
| Dimensions | 122 × 75 × 26.4 mm (4.80 × 2.95 × 1.04 in) |
| Weight | 235 g (0.52 lb) |
| Housing | Plastic |
| Cooling | Fanless |
| Operating Temperature | -10 °C ~ +50 °C (14 °F ~ +122 °F) |
| Storage Temperature | -20 °C ~ +60 °C (-4 °F ~ +140 °F) |
| Protection | IP30 |
| Certification | FCC, IC, PTCRB, Verizon, AT&T,<br>FirstNet*, T-MOBILE, CE, UN38.3 |
| Warranty | 3 years (Battery: 1 year) |

## Software

| | |
|---|---|
| Network Access | APN authentication |
| Authentication | CHAP/PAP |
| Network Functions | Network Standard; LAN Protocol; WAN Protocol; IP Application; Routing; NAT |
| Firewall | SPI, DoS protection |
| Access Control | ACL, URL filtering |
| Link Backup | Hot backup, cold backup, load balancing |
| Dual-SIM Failover | Supports dual-SIM failover (North American model only) |
| Reliability | Link Detection; Watchdog |
| Configuration | Telnet, Web, SSH |
| Upgrade | Web upgrade, Device Manager upgrade |
| Operations and Management | SMS Functions; Dial-on-demand; Traffic Management; Alarm; Maintenance Tools; Status Query; Device Manager |

# Selection Guide

| | Model | CR602 | CR202-Lite |
|---|---|---|---|
| Performance | Access capacity | Up to 32 Wi-Fi devices; up to 96 wired devices via switch expansion | Up to 32 users |
| Interface | Cellular | 5G Release 16; up to 7.01 Gbps DL | 4G LTE Cat.4/Cat.6; up to 300 Mbps DL |
| | Ethernet | 2 × 2.5GbE | 2 × Fast Ethernet |
| | Wi-Fi | Wi-Fi 7 BE3000; dual-band 3 × 3 MIMO | 2.4 GHz Wi-Fi 4; up to 300 Mbps |
| | SIM | 2 x Nano SIM + 1 x eSIM optional | Nano-SIM + optional eSIM |
| | USB-C | USB-C charging and firmware upgrade | USB-C power input |
| | Antenna configuration | 2 × external + 6 × built-in cellular; 3 × built-in Wi-Fi | 2 × external cellular + 2 × internal Wi-Fi |
| Power | Battery | 10,000 mAh | Optional 3,000 mAh |
| | Maximum stated runtime | Up to 15 hours in 5G mode | Up to 8 hours |
| | Power input | USB-C or battery power | USB-C, 5 V / 2 A |
| Mechanical | Dimensions | 167 × 102 × 36.5 mm | 122 × 75 × 26.4 mm |
| | Weight | Approx. 550 g with battery | 235 g |
| Environment | Protection rating | IP30 | IP30 |
| | Operating temperature | -10 °C to +40 °C | -10 °C to +50 °C |
| | Operating humidity | 5–95% RH, non-condensing | 5–95% RH, non-condensing |
| Software | Cloud management | InCloud Manager | Device Manager |
| | Link continuity | Primary/backup or load balancing; link status, latency, packet-loss and jitter monitoring | Hot/cold backup and load balancing; dual-SIM failover on North American model |
| | Security | Firewall, IPsec, L2TP, policy routing, traffic shaping | SPI firewall, DoS protection, ACL, URL filtering |
| Certification | Certification | FCC, IC, PTCRB, RoHS, REACH, Verizon, AT&T, T-Mobile, FirstNet, CE* UN38.3 | FCC, IC, PTCRB, Verizon, AT&T, FirstNet*, T-Mobile, CE, UN38.3 |



# Vehicle Routers

## VR624

Rugged in-vehicle connectivity for public transport, service fleets, and mobile operations. VR624 combines 5G, Wi-Fi 6, M12 Gigabit Ethernet, GNSS positioning, and centralized fleet management.




# VR624

## Rugged 5G Vehicle Router

### 5G and Wi-Fi 6

High-speed links support onboard video, telemetry, and passenger applications.

### Four M12 ports

Vibration-resistant Gigabit connectors secure cameras, computers, and onboard systems.

### Integrated GNSS

Five satellite constellations support fleet positioning and route-aware applications.

### Protected fleet operations

TPM 2.0, VPN, firewall, and DeviceLive support secure remote maintenance.

VR624 brings 5G, Wi-Fi 6, rugged M12 Ethernet, GNSS positioning, and cloud management to vehicles. Its protected power input and fanless metal enclosure suit public transport, emergency, service, and logistics fleets.

## Hardware

| | |
|---|---|
| Processor | 1.3 GHz industrial-grade CPU |
| Memory / Storage | 512 MB DDR3 / 128 MB Flash |
| Cellular | 5G NR SA/NSA; theoretical downlink up to 3.4 Gbps |
| SIM / eSIM | Dual Nano-SIM drawer; optional eSIM |
| Ethernet | 4 × M12 X-code Gigabit ports |
| Digital Input | 2 × input for ignition or low-battery detection |
| Wi-Fi | Wi-Fi 6 AX3000, dual-band |
| Wi-Fi Security | WPA3-Enterprise, WPA2-Enterprise, WEP/TKIP/AES |
| GNSS | GPS, BDS, Galileo, GLONASS, QZSS |
| Positioning | <1.5 m CEP50; 10 Hz update rate |
| Antenna Interfaces | 4 × 5G SMA; 2 × 4G SMA; 2 × Wi-Fi RP-SMA; 1 × GNSS SMA |
| Indicators | System, cellular, signal, 2.4 GHz Wi-Fi, 5 GHz Wi-Fi |
| Security / RTC | TPM 2.0 / real-time clock |

| | |
|---|---|
| Power Input | 9–48 VDC; over-current and reverse-polarity protection |
| Power Consumption | 5.3 W standby; 8.5 W typical; 17.8 W maximum |
| Enclosure / Cooling | Metal / fanless passive cooling |
| Protection | IP30 |
| Operating Temperature | -30°C ~ +70°C (-22 °F~ +158 °F) |
| Storage Temperature | -40°C ~ +85°C (-40 °F~ +185 °F) |
| Humidity | 5–95%, non-condensing |
| Dimensions / Weight | 177 × 144.4 × 42 mm / 810 g |
| Ground Terminal | Supported |
| Rail Standard EMC | EN50155, EN50121 |
| Fire Prevention | EN45545 |
| Certification | CE, E-Mark, ITxPT*, FCC*, IC*, PTCRB*, Verizon*, AT&T*, T-Mobile* |

## Software

| | |
|---|---|
| Network Access | APN, VPDN; CHAP/PAP authentication |
| LAN / WAN | Ethernet, routing, DHCP, DNS, DDNS |
| Firewall | SPI, rules, forwarding, SNAT/DNAT, MAC/domain filtering |
| Access Control | 802.1X and MAC/IP/port/protocol policies |
| VPN / Security | VPN support, TPM 2.0, policy routing, QoS |
| Link Management | WAN/cellular failover, standby/load balancing, ICMP detection |
| Dual-SIM Recovery | Policy-based switching, heartbeat, auto-redial, watchdog |
| Wi-Fi Modes | AP/client modes, WPA/WPA2/WPA3, portal authentication |
| DTU | TCP/UDP transparent transmission; Modbus RTU-to-TCP conversion |
| Fleet Management | Batch configuration, monitoring, OTA updates, alerts, location |
| Diagnostics | Logs, alarms, Ping, traceroute, packet capture, iPerf |
| System Management | Web/CLI access, configuration backup, firmware upgrades |
| Positioning Service | GNSS forwarding, tracking, geofencing, route history |

# Industrial Routers




**IR302 / IR315 / IR602 / IR624**

Industrial routers for distributed equipment, unattended sites, and harsh environments. The portfolio ranges from compact 4G models to 5G RedCap and full 5G platforms, with industrial interfaces, link resilience, secure tunneling, and remote management.



# IR925

## 5G Industrial Router with 2.5GbE

### 5G and Wi-Fi 6

Supports 5G downlink rates up to 3.4 Gbps and dual-band AX3000 Wi-Fi 6.

### Multi-Gigabit Connectivity

One 2.5GbE port, four Gigabit ports, serial interfaces, DIO, and relay connect field equipment.

### Resilient Connectivity

Dual-SIM failover, link detection, automatic reconnection, and an embedded watchdog support unattended sites.

### Secure cloud management

TPM, VPN, firewall, and DeviceLive support secure connectivity, centralized monitoring, and remote maintenance.

IR925 combines 5G, Wi-Fi 6, multi-gigabit Ethernet, and industrial I/O for demanding field applications, with resilient link management and DeviceLive support for remote operations.

## Hardware

| | |
|---|---|
| Processor | 1 GHz |
| Memory | 512 MB |
| Storage | 8 GB eMMC |
| Cellular | 5G NR, downlink up to 3.4 Gbps,<br>Sub-6 GHz (450 MHz-6 GHz);<br>backward-compatible with 4G/3G |
| SIM | 2 x Nano SIM; 1 x eSIM optional |
| Ethernet | 1 × 2.5GbE + 4 × 1GbE RJ45;<br>WAN/LAN/VLAN configurable; 1.5 kV isolation |
| Serial Ports | 1 × RS232 and 1 × RS485; 15 kV ESD protection |
| Digital I/O and Relay | 4 × DIO (DI/DO configurable);<br>1 × relay, 2 A @ 30 V DC |
| Wi-Fi | Dual-band 2.4 GHz and 5.8 GHz Wi-Fi 6;<br>up to 3000 Mbps; AP/Client modes |
| Antenna Connectors | 4 × 5G or 2 × 4G; 2 × Wi-Fi; 1 × GNSS |

| | |
|---|---|
| Indicators | System, cellular, 2.4 GHz Wi-Fi, 5 GHz Wi-Fi, and signal |
| Hardware Security | TPM supported; RTC supported |
| Power Input | 9-48 V DC; overcurrent and reverse-polarity protection |
| Enclosure and Cooling | Metal enclosure; fanless cooling |
| Protection Rating | IP30 |
| Operating Temperature | -40 °C ~ +70 °C（-40°F ~ +158°F） |
| Storage Temperature | -40 °C ~ +85 °C（-40°F ~ +185°F） |
| Humidity | 5%-95% RH, non-condensing |
| Dimensions | 141.2 × 118.5 × 53 mm |
| Weight | 1002 g |
| Mounting | DIN rail or wall mount |
| Physical Tests | IEC 60068-2-27 shock; IEC 60068-2-6 vibration;<br>IEC 60068-2-32 free fall |
| EMC Immunity | EN 61000-4 series, mainly Level 3 |

## Software

| | |
|---|---|
| Network Access | APN, VPDN; CHAP/PAP authentication |
| WAN Protocols | Static IP, DHCP, PPPoE |
| Routing | Static routing, OSPF, BGP |
| Network Services | IPv4/IPv6, TCP/UDP, NTP, DNS, DHCP Server/Relay/Client, DDNS, IP passthrough |
| Firewall | MAC/IP/port/protocol filtering, NAT, port mapping, access control, policy-based routing, 802.1X |
| VPN and Certificates | IPsec, L2TP, OpenVPN, WireGuard; CA certificate support |
| Link Backup | Interface backup and dual-SIM failover |
| Link Detection | Heartbeat-based detection with automatic reconnection |
| Watchdog | Embedded watchdog with automatic fault recovery |
| Wi-Fi Features | AP/Client modes, captive portal, multi-SSID, WPA/WPA2/WPA3 |
| GNSS | IP forwarding, serial forwarding, and time synchronization |
| DTU and Modbus | TCP/UDP transparent transmission; Modbus RTU-to-TCP conversion |
| Monitoring | Dashboard, link-quality monitoring, traffic statistics, and cellular signal metrics |
| Logs and Alarms | System and diagnostic logs, device events, and email alarms |
| Management | Web/CLI access, SNMP v1/v2c/v3, configuration import/export, and network diagnostics |
| Cloud Management | DeviceLive centralized management, monitoring, remote maintenance, and remote connectivity |



# IR624

## Multi-Port 5G Industrial Router

### Full 5G connectivity

5G SA/NSA model options support theoretical downlink rates up to 3.4 Gbps with LTE fallback.

### Rich industrial interfaces

Four isolated Gigabit Ethernet ports, RS-232, and RS-485 connect cameras, controllers, and other field equipment.

### Resilient unattended operation

Dual-SIM failover, link detection, interface backup, an embedded watchdog, and wide-voltage input support remote sites.

### Secure cloud-managed networking

DeviceLive, VPN, firewall, policy routing, and 802.1X support protected connectivity and centralized maintenance.

IR624 is a full 5G industrial router for sites with higher bandwidth and interface requirements. Four Gigabit ports, serial connectivity, optional dual-band Wi-Fi, secure tunneling, and DeviceLive support video, automation, and remote operations.

## Hardware

| | |
|---|---|
| CPU | 880 MHz |
| RAM | 256 MB |
| Flash | 128 MB |
| Ethernet Ports | 4 x 10/100/1000 Mbps Ethernet (WAN/LAN/VLAN), 1.5 kV isolation |
| Power Interface | DC 9 to 48V with over-current and reverse-polarity protection |
| Serial Port | 1 x RS-232 and 1 x RS-485; ESD protection: 15KV |
| Reset Button | Supported |
| SIM Slot | 2 x Nano SIM; 1 x eSIM optional |
| Antenna Connectors | 4 x SMA for 5G or 2 x SMA for 4G, 2 x RP-SMA for Wi-Fi |
| Wi-Fi Frequency | 2.4 GHz / 5 GHz |
| Wi-Fi Max Bandwidth | Up to 300 Mbps (2.4G) / 867 Mbps (5G) |
| Wi-Fi Transfer Protocol | IEEE 802.11 ac/a/b/g/n Wave2 MU-MIMO |
| Wi-Fi Transmit Power | 5G:17 dBm; 2.4G:17 dBm |
| Wi-Fi Transmission Distance | 50 meters by line of sight(Actual transmission distance depends on environment of the site.) |
| Standby Power | 370-480mA@12V |
| Working Power | 415-530mA@12V |
| Peak Power | 530mA@12V |
| Product Dimensions | 127 × 108.2 × 35 mm (5 × 4.26 × 1.38 in) |
| Product Weight | 544 g (1.2 lb) |
| Mounting Method | DIN rail or wall |
| Protection Rating | IP30 |
| Housing and Cooling | Fanless metal housing |
| Storage Temperature | -40 °C ~ 85 °C (-40 °F ~ 185 °F) |
| Operating Temperature | Normal: -35 °C ~ +70 °C (-31 °F ~ +158 °F); Extended:-40 °C ~ +75 °C (-40 °F ~ +167 °F) |
| Ambient Humidity | 5 to 95% RH, non-condensing |
| Physical Characteristics | IEC60068-2-27 Shockproof; IEC60068-2-6 Vibration Resistance; IEC60068-2-32 Free Fall |
| EMC Standard | EN61000-4-2, level 3, Static; EN61000-4-3, level 3, Radiation Electric Field; EN61000-4-4, level 3, Pulsed Electric Field; EN61000-4-5, level 3, Surge; EN61000-4-6, level 3, Conducted Distubance Immunity; EN61000-4-8, Power Frequency Field Resistance, horizontal / vertical 400A/m (>level 2); EN61000-4-12, level 3, Shock Wave Resistance |
| Certifications | CE, E-MARK, ECE R118, FCC, IC, PTCRB, AT&T, T-Mobile, Verizon(NSA) |

## Software

| | |
|---|---|
| Network Access | APN/VPDN |
| Access Authentication | CHAP/PAP |
| Network Functions | Network Type; LAN protocol; WAN protocol; IP Applications; IP Routing; NAT Functions |
| Network Security | Access control, firewall filtering (MAC/IP/port/protocol), policy-based routing, 802.1X |
| Data Security | IPsec VPN, L2TP VPN, OpenVPN, WireGuard VPN |
| Certificate Management | CA Certificates |
| Link Detection | Heartbeat link detection with auto-redial |
| Embedded Watchdog | Supported |
| Link Backup and Failover | Backup; Dual-SIM Switching |
| Operating Mode | AP/Client modes |
| Security Features | WPA/WPA2, WEP/TKIP/AES, Wi-Fi Portal support |
| DTU Function | TCP/UDP transparent mode |
| Bridge | Modbus RTU-to-TCP bridge |
| QoS Management | QoS traffic shaping |
| Configuration Methods | Web/CLI remote access, config import/export |
| Operations and Monitoring | Log & Alarm; Network Management Function; Maintenance Tools; Monitoring |




# IR602

## 5G RedCap Industrial Router

### 5G RedCap for industrial IoT

The global model supports theoretical peak rates up to 223 Mbps downlink and 123 Mbps uplink on 5G SA.

### Fast local connectivity

Wi-Fi 6 AX3000, one 2.5GbE port, and one Gigabit port provide high-speed access for local terminals.

### Industrial resilience

Dual Nano-SIM, link scheduling, watchdog functions, GPIO, a metal enclosure, and wide-temperature operation support unattended sites.

### Centralized industrial operations

DeviceLive supports batch configuration, remote upgrades, diagnostics, alerts, and remote maintenance.

IR602 brings 5G RedCap, Wi-Fi 6, 2.5GbE, and industrial I/O to distributed industrial sites. It targets mid-bandwidth applications that need 5G SA evolution, compact hardware, secure remote connectivity, and fleet-level operations.

## Hardware

| | |
|---|---|
| Processor | Dual-core Cortex-A53, 1.3 GHz |
| Memory / Storage | 512 MB DDR3 / 128 MB SPI NAND Flash |
| Cellular | 5G NR SA; 4G LTE FDD/TDD |
| Peak Cellular Rate | Global model: up to 223 Mbps DL / 123 Mbps UL, theoretical |
| SIM / eSIM | 2 x Nano SIM; 1 x eSIM optional |
| Ethernet | 1 × 2.5GbE WAN/LAN; 1 × GbE LAN; 1.5 kV isolation |
| Industrial I/O | 2 × GPIO; DI: 0–30 V; DO: open-drain output |
| Wi-Fi | Wi-Fi 6 AX3000, dual-band |
| Wi-Fi Security | WPA3-Enterprise, WPA2-Enterprise, WEP/TKIP/AES |
| Wi-Fi Transmit Power | 20 dBm at 2.4 GHz and 5 GHz |
| Antenna Connectors | 4 × SMA: 2 cellular + 2 Wi-Fi |
| Indicators / Reset | System, network, signal, dual-band Wi-Fi / pinhole reset |
| Power Input / Consumption | 12 VDC / 2 A; ≤15 W |
| Enclosure / Cooling | Metal / fanless passive cooling |
| Protection | IP30 |
| Operating Temperature | -40°C~ +70°C（-40 °F~ +158 °F）extended: ~ +75°C（~ +167 °F） |
| Storage Temperature | -40°C~ +85°C（-40 °F~ +185 °F） |
| Humidity | 5–95%, non-condensing |
| Dimensions / Weight | 98.3 × 92 × 24 mm / approx. 390 g |
| Mounting | DIN rail or wall |
| Shock / Vibration / Free Fall | IEC 60068-2-27 / -6 / -32 |
| EMC Immunity | EN 61000-4-2/-3/-4/-8, Level 3 |
| Certifications & Compliance | CE, FCC, PTCRB, IC, RoHS, REACH, Verizon, AT&T, T-Mobile |

## Software

| | |
|---|---|
| Cellular Access | 5G NR SA, 4G LTE; APN, VPDN, dual APN |
| Dual-SIM | Dual Nano-SIM with automatic hot backup on dial failure |
| WAN / LAN | DHCP, static IP, PPPoE; 1 × GbE LAN |
| Wi-Fi | 802.11 ax/ac/a/b/g/n; WPA/WPA2/WPA3-PSK |
| Multi-WAN | Ethernet WAN, cellular, and Wi-Fi STA |
| Link Backup | Priority-based backup, heartbeat detection, automatic reconnection |
| Reliability | Built-in watchdog with automatic fault recovery |
| Routing / NAT | Static routing, BGP, OSPF, SNAT/DNAT, policy routing |
| VPN | IPsec IKEv1/IKEv2 and OpenVPN client/server |
| Traffic Management | Interface bandwidth control and QoS queue management |
| LAN Services | DHCP server/client, DNS proxy/filtering, DDNS |
| Bridge Mode | WAN or cellular IP passthrough |
| Monitoring | Device status, signal, data usage, and link quality |
| Diagnostics | Ping, traceroute, packet capture, and iPerf |
| Management | Web UI and SSH CLI |
| Logs / Access Control | System, diagnostic, and event logs; role-based permissions |




# IR302

## Compact Industrial Cellular Router

### Flexible 4G access

Dual-SIM LTE configurations connect distributed industrial and commercial equipment.

### Industrial interfaces

Two Fast Ethernet ports and RS-232/RS-485 connect common field devices.

### Link resilience

Dual-SIM switching, link detection, auto-redial, and watchdog support unattended sites.

### Secure remote access

VPN, firewall, Device Manager, and InConnect support remote maintenance.

IR302 is a compact industrial router for equipment connectivity, data backhaul, and remote maintenance. It combines cellular access, wired Ethernet, serial communication, VPN, link resilience, and cloud management in a fanless metal enclosure.

## Hardware

| | |
|---|---|
| Processor | 580 MHz industrial-grade processor |
| Memory / Storage | 128 MB DDR / 32 MB SPI Flash |
| 4G Module | Model-dependent industrial LTE module; see Ordering Information |
| Ethernet | 1 × 10/100 Mbps WAN/LAN; 1 × 10/100 Mbps LAN |
| Serial Port | 1 × RS-232/RS-485 combo; DTU transparent transmission |
| SIM | 1 x Nano SIM |
| Antennas | 2 × cellular SMA; 1 × Wi-Fi SMA |
| Power Input | 9–36 VDC terminal; optional 12 V adapter |
| Power Consumption | Typical 2–5 W; peak <10 W |
| Indicators / Reset | Power, status, cellular, signal, Wi-Fi / factory reset |
| Enclosure / Protection | Metal, IP30 |
| Mounting | DIN rail or wall |
| Dimensions / Weight | Approx. 100 × 30 × 100 mm / 250 g |

| | |
|---|---|
| Operating Temperature | Normal: -35°C~ +70°C（-31 °F~ +158 °F） extended: -40°C~ +75°C（-40 °F~ +167 °F） |
| Storage Temperature | -40°C~ +85°C（-40 °F~ +185 °F） |
| Humidity | 5–95% RH, non-condensing |
| Power Protection | Reverse-polarity protection |
| Surge / ESD | Surge: 2 kV common, 1 kV differential; ESD: 4/8 kV |
| Certifications | CE, CB, UKCA, E-MARK, FCC, IC, PTCRB, AT&T, Verizon, T-Mobile, RCM, CCC, EAC&FAC, UL, ANATEL, MIC&JATE,IEC62443, NOM, IFETEL, UL/cUL, CB (IEC 62368-1) |
| EMC | IEC 61000-4-2/-3/-4/-5/-6/-8 |
| Warranty | 3 years |

## Software

| | |
|---|---|
| Wi-Fi | 1×1 AP; up to 72 Mbps; client isolation and MAC filtering |
| WAN / LAN | PPPoE, DHCP Client, static IP; 10/100 Mbps |
| DHCP / DNS | DHCP server/relay; DNS forwarding and custom DNS |
| Routing / NAT | Static routing, NAT, DDNS, NTP |
| VPN | IPsec, OpenVPN, GRE, L2TP, PPTP, WireGuard, ZeroTier |
| Firewall / Access Control | ACLs, MAC-IP binding, filtering, attack protection |
| Link Backup | 4G, wired WAN, and Wi-Fi STA switching; VRRP and load balancing |
| Dual-SIM Management | Signal, packet-loss, dial-attempt, and connection-time policies |
| Serial / Ethernet DTU | TCP/UDP/MQTT transparent transmission; Modbus RTU-to-TCP conversion |
| Cloud Management | Device Manager and InConnect |
| SNMP | v1/v2c/v3, MIB-II/private MIBs, Trap alarms |
| Monitoring / Diagnostics | Status, traffic, logs, alarms, Ping, traceroute, packet capture |
| Management Access | WebUI, CLI, SSH, HTTP API |
| Traffic Management | IP-based rate limiting, statistics, and traffic alarms |
| Maintenance | Scheduled reboot, scheduled cellular redial, firmware upgrades |
| Industrial Functions | Smart ATM, Smart-EMS, and State Grid encryption support |



# IR315

## Economical Multi-Port Industrial Router

### Five-port industrial networking

Five Fast Ethernet ports support configurable WAN, LAN, and VLAN designs for multiple field devices.

### Serial or digital I/O options

Model options provide RS-232 plus RS-485 or four digital I/O channels for equipment integration.

### High-availability functions

Dual-SIM policies, link detection, VRRP, load balancing, and watchdog functions support resilient site connectivity.

### Broad protocol and cloud support

Multiple VPN protocols, SNMP, Device Manager, and InConnect support secure networking and remote operations.

IR315 expands industrial connectivity with five Ethernet ports and optional serial or I/O configurations. It is suited to equipment-dense sites that need flexible local networking, cellular resilience, secure tunnels, and centralized management.

## Hardware

| | |
|---|---|
| CPU | 580 MHz |
| Memory / Storage | 128 MB DDR2 / 64 MB SPI Flash |
| Ethernet | 5 × Fast Ethernet; WAN/LAN/VLAN; 1.5 kV isolation |
| Power Input | 9–36 VDC terminal; over-current and reverse-polarity protection |
| Optional Interfaces | 4 × configurable DI/DO or 1 × RS-232 + 1 × RS-485 |
| SIM | 1 x Nano SIM |
| Antenna Connectors | 2 × cellular SMA; 2 × Wi-Fi RP-SMA |
| GNSS | Optional on selected models; see Ordering Information |
| Wi-Fi | 802.11 b/g/n, 2.4 GHz, up to 300 Mbps |
| Wi-Fi Modes | AP, client, and WDS |
| Wi-Fi Range | Approx. 50 m line of sight; environment dependent |
| Indicators / Reset / Ground | Power, system, network, Wi-Fi, signal / pinhole reset / grounding |
| Power Consumption | 120–200 mA standby; 150–320 mA operating at 12 V |
| Dimensions / Weight | 127 × 108.2 × 35 mm / 454 g |
| Mounting | DIN rail or wall |
| Enclosure / Cooling | Metal / fanless |
| Protection | IP30 |
| Operating Temperature | Normal: -35°C~ +70°C（-31 °F~ +158 °F） extended: -40°C~ +75°C（-40 °F~ +167 °F） |
| Storage Temperature | -40°C~ +85°C（-40 °F~ +185 °F） |
| Humidity | 5–95%, non-condensing |
| Shock / Vibration / Drop | IEC 60068-2-27 / -6 / -32 |
| EMC Immunity | EN 61000-4-2/-3/-4/-5/-6/-8/-12 |
| Certifications | CE, E-MARK, FCC, IC, PTCRB, AT&T, Verizon, T-Mobile, RCM, IMDA, MIC/JATE, SRRC, UL, C1D2, IEC 62443-4-2, EN 18031 |

## Software

| | |
|---|---|
| Cellular Access | Model-dependent 2G/3G/4G/5G standards; IPv4/IPv4v6 |
| SIM Management | Dialing, dual-SIM switching, operator settings, signal reporting |
| WAN / VLAN | Static IP, DHCP, PPPoE; access/trunk VLAN |
| Ethernet | Port roles, isolation, ARP, and IP passthrough |
| Wi-Fi | 802.11 b/g/n; WEP, WPA/WPA2-PSK |
| Link Backup | Cellular/WAN backup, hot/cold standby, and load balancing |
| High Availability | VRRP with virtual MAC and priority election |
| Routing / NAT | Static routing, OSPF, DDNS, NAT, port and virtual-IP mapping |
| VPN | IPsec, DMVPN, PPTP, L2TP, GRE, OpenVPN, WireGuard, ZeroTier |
| Firewall / Access Control | SPI, DoS protection, ACL, filtering, DMZ, IP-MAC binding, 802.1X |
| SNMP | v1/v2c/v3 with Trap alarms |
| Management | Device Manager, InConnect, Web, API, SSH/Telnet, console |
| Logs / Alarms | Local and remote logs; traffic, system, status, and SMS alarms |
| DTU / Serial | TCP/UDP transmission, Modbus bridge, serial traffic statistics |
| Industrial Functions | State Grid encryption, I/O control, SMART-EMS |
| LAN / Maintenance | DHCP, DNS, NTP, portal, QoS, scheduled tasks, packet capture |

# Selection Guide

| | Model | IR925 | IR624 | IR602 | IR302 | IR315 |
|---|---|---|---|---|---|---|
| Hardware platform | CPU | 1 GHz | 880 MHz | Dual-core Cortex-A53 @ 1.3 GHz | 580 MHz | 580 MHz |
| | RAM | 512 MB | 256 MB | 512 MB DDR3 | 128 MB | 128 MB DDR2 |
| | Flash / storage | 8 GB eMMC | 128 MB | 128 MB SPI NAND Flash | 32 MB SPI Flash | 64 MB SPI |
| Interface | Ethernet | 1 × 2.5GbE + 4 × GbE | 4 × GbE | 1 × 2.5GbE + 1 × GbE | 2 × Fast Ethernet | 5 × Fast Ethernet |
| | Serial / I/O | 1 × RS232 + 1 × RS485; 4 × DIO + 1 × relay | 1 × RS-232 + 1 × RS-485 | 2 × GPIO | 1 × RS-232/RS-485 combo | 1 × RS-232 + 1 × RS-485 or 4 × I/O |
| | Cellular | 5G SA/NSA + 4G/3G fallback | 5G SA/NSA or LTE model options | 5G RedCap SA + LTE | 4G LTE model options | 4G LTE model options |
| | Wi-Fi | Wi-Fi 6 AX3000 | Wi-Fi 5 AC1200 | Wi-Fi 6 AX3000 | Wi-Fi 4 | Wi-Fi 4 |
| | SIM | 2 x Nano SIM; 1 x eSIM optional | 2 x Nano SIM; 1 x eSIM optional | 2 x Nano SIM; 1 x eSIM optional | 1 x Nano SIM | 1 x Nano SIM |
| Power | Power input | 9-48 V DC | 9–48 V DC | 9-48V DC | 9–36 V DC | 9–36 V DC |
| Mechanical | Dimensions | 141.2 × 118.5 × 53 mm | 127 × 108.2 × 35 mm | 98.3 × 92 × 24 mm | Approx. 100 × 30 × 100 mm | 127 × 108.2 × 35 mm |
| | Weight | Approx.1002 g | Approx.544 g | Approx. 390 g | Approx. 250 g | Approx.454 g |
| | Mounting | DIN rail / wall mount | DIN rail / wall mount | DIN rail / wall mount | DIN rail / wall mount | DIN rail / wall mount |
| | Housing | Metal, fanless | Metal, fanless | Metal, fanless | Metal, fanless | Metal, fanless |
| | Protection rating | IP30 | IP30 | IP30 | IP30 | IP30 |
| Environment | Operating temperature | Standard -40 °C to +70 °C; extended +75°C | Standard -35 °C to +70 °C; extended -40 °C to +75 °C | Standard -40 °C to +70 °C; extended +75 °C | Standard -35 °C to +70 °C; extended -40 °C to +75 °C | Standard -35 °C to +70 °C; extended -40 °C to +75 °C |
| | Humidity | 5%-95% RH, non-condensing | 5–95% RH, non-condensing | 5–95%, non-condensing | 5–95% RH, non-condensing | 5–95%, non-condensing |
| Reliability | Physical tests | IEC 60068-2-27 / -2-6 / -2-32 | IEC 60068-2-27 / -2-6 / -2-32 | IEC 60068-2-27 / -2-6 / -2-32 | IEC 60068-2-27 / -2-6 / -2-32 | IEC 60068-2-27 / -2-6 / -2-32 |
| | EMC | EN 61000-4 series, mainly Level 3 | EN 61000-4 series, mainly Level 3 | EN 61000-4 series, mainly Level 3 | IEN 61000-4 series, mainly Level 2 | EN 61000-4 series, mainly Level 2 |
| Software | Cloud management | DeviceLive | DeviceLive | DeviceLive | Device Manager / InConnect | Device Manager / InConnect |
| | VPN | IPsec, L2TP, OpenVPN, WireGuard | IPsec, L2TP, OpenVPN, WireGuard | IPsec, L2TP, GRE, OpenVPN, WireGuard, ZeroTier | IPsec, GRE, L2TP, PPTP, OpenVPN, WireGuard, ZeroTier | IPsec, PPTP, L2TP, GRE, DMVPN, OpenVPN, WireGuard, ZeroTier |
| Certification | Certification | CE*, FCC*, PTCRB*, IC*, RoHS, REACH, Verizon*, AT&T*, T-Mobile* | CE, E-Mark, ECE R118, FCC, IC, PTCRB, AT&T, T-Mobile, Verizon (NSA) | CE*, FCC*, PTCRB*, IC*, RoHS, REACH, Verizon*, AT&T*, T-Mobile* | CE, CB, UKCA, E-MARK, FCC, IC, PTCRB, AT&T, Verizon, T-Mobile, RCM, CCC, EAC&FAC, UL, ANATEL, MIC&JATE,IEC62443, NOM, IFETEL, UL/cUL, CB (IEC 62368-1) | CE, E-Mark, FCC, IC, PTCRB, AT&T, Verizon, T-Mobile, RCM, IMDA, MIC/JATE, SRRC, UL, C1D2, IEC 62443-4-2, EN 18031 |

# Cloud Services



## InCloud Manager / DeviceLive / Device Manager / InConnect

Cloud services for network operations, industrial device management, remote maintenance, and secure access. Each service has a distinct role and is presented separately to avoid a misleading feature-by-feature comparison.



# InCloud Manager

## Cloud-Managed Networking Platform

### Centralized network management

Manage routers, access points, and switches with shared visibility, remote configuration, upgrades, and alerts.

### AI-assisted device operations

Natural-language requests can retrieve information, analyze issues, prepare actions, and guide approved operational steps.

### Controlled execution

Permissions, user confirmation for write operations, and activity logs keep operational changes visible and accountable.

### Branch connectivity and remote access

SD-WAN, remote maintenance, and mobile access support distributed stores, offices, and service locations.

InCloud Manager brings deployment, monitoring, maintenance, and branch networking into one cloud platform. Its AI-assisted workflow works within user permissions, requests confirmation before write operations, and records activity for review.

# Service / Software Specifications

| | |
|---|---|
| Platform Scope | Cloud management for supported routers, access points, and switches |
| AI Network Assistant | Uses natural-language requests for selected monitoring, diagnosis, and device operations |
| Device Monitoring | Tracks online status, traffic, connected clients, and online history |
| Fault Diagnosis | Reviews device status, signal trends, and logs to support troubleshooting |
| Signal Analysis | Evaluates current and historical cellular signal quality and identifies weak-signal devices |
| Remote Diagnostics | Supports Ping, traceroute, packet capture, and speed testing |
| Configuration Management | Supports configuration reading, copying, backup, restore, and confirmed changes |
| Firmware Management | Supports compatibility checks, staged upgrades, progress tracking, and rollback |
| Alert Management | Displays, acknowledges, and manages device and network alerts |
| Remote Access | Opens supported device Web interfaces and command-line sessions |
| Batch Operations | Supports batch inspection, configuration, firmware, and license management |
| Access Control and Audit | Applies user permissions and confirmation before write operations, with activity logging |



# DeviceLive

## Industrial IoT Device Management Platform

### Unified device management

Onboard, group, configure, upgrade, and monitor supported InHand industrial devices from a central platform.

### Monitoring and diagnostics

Dashboards, alert policies, signal and network-quality data, logs, and diagnostic tools support remote troubleshooting.

### Edge application management

Deploy and maintain supported edge applications without building a separate over-the-air update service.

### Remote terminal maintenance

Connector creates on-demand access to downstream PLCs, HMIs, cameras, and computers for authorized maintenance.

DeviceLive supports the operational lifecycle of industrial routers, gateways, and edge computers. It combines device management, monitoring, diagnostics, edge application management, and controlled remote access to downstream equipment.

# Service / Software Specifications

| | |
|---|---|
| Device Onboarding | Supports cloud-based device onboarding and remote deployment |
| Device Groups | Organizes devices by site, customer, or business need |
| Configuration Management | Supports remote and batch configuration delivery |
| Firmware Management | Supports scheduled and batch firmware upgrades |
| Device Monitoring | Tracks connection status, network type, and device health |
| Network Monitoring | Tracks interface status, links, traffic, and cellular signal |
| Network Quality | Monitors latency, jitter, packet loss, and throughput |
| Remote Diagnostics | Provides diagnostic logs, Ping, traceroute, packet capture, and event analysis |
| Alerts | Supports configurable alerts with SMS, email, and app notifications |
| Geolocation | Supports GPS, base-station, or manual positioning with map views |
| DSA Management | Supports remote DSA configuration, upgrades, and status monitoring |
| Remote Terminal | Provides authorized remote access to supported downstream devices (Advanced Edition) |
| Edge App Management | Manages supported containers and edge applications (Advanced Edition) |
| Security and Mobile App | Supports role-based access, MFA, QR-code onboarding, and mobile monitoring |



# Device Manager

## Cloud Device Management Platform

### Device lifecycle management

Manage device information, firmware, configuration, logs, and supported edge applications from one portal.

### Remote configuration and upgrades

Distribute configuration tasks, schedule firmware upgrades, reboot devices, and open supported remote web interfaces.

### Fleet monitoring and alerts

Dashboards, GIS views, traffic analysis, signal data, task tracking, and event alerts provide operational visibility.

### Permissions and integration

User roles, access controls, system logs, MQTT-based connectivity, and integration services support controlled operations.

Device Manager is a cloud platform for managing InHand industrial routers and gateways across distributed sites. It focuses on device inventory, remote configuration, firmware, monitoring, alerts, role-based access, and operational records.

# Service / Software Specifications

| | |
|---|---|
| Device Inventory | Manages serial number, model, firmware, configuration state, signal, and cellular identifiers |
| Configuration Management | Supports batch configuration, task delivery, and remote reboot |
| Firmware Management | Supports batch firmware upgrades and upgrade-task tracking |
| Remote Web Access | Opens supported router Web interfaces for remote configuration |
| Edge App Management | Installs and updates supported edge applications and configurations |
| Device Monitoring | Analyzes traffic, online status, and cellular signal quality |
| Logs | Provides device-log viewing and system-log tracking |
| Event Alerts | Reports device, traffic, SIM, link, and port-status events |
| Location Management | Displays and manages device locations on GIS maps |
| Task Tracking | Tracks device-management task status and progress |
| Users and Permissions | Manages accounts, roles, and device access permissions |
| Integration | Supports integration with third-party services |




# InConnect

## Secure Remote Access and Site Connectivity Service

### Remote access to field devices

Authorized users can reach downstream PLCs, HMIs, industrial PCs, cameras, servers, and controllers without traveling onsite.

### Multiple connectivity modes

User-to-device, peer-to-peer, and star-network options support maintenance, remote work, and site interconnection.

### Identity and access controls

User accounts, access duration, network groups, and device permissions limit who can reach each resource.

### Scalable site connectivity

Centralized account and site management supports large distributed deployments with controlled client access.

InConnect provides secure remote access and connectivity for distributed industrial and enterprise sites. It gives authorized users controlled paths to routers, gateways, and downstream equipment for maintenance, program deployment, and data access.

# Service / Software Specifications

| | |
|---|---|
| Router and Gateway Management | Manages device identity, network status, traffic, and signal history |
| Terminal Device Management | Manages and provides remote access to supported PLCs, HMIs, and other downstream devices |
| Configuration Management | Supports batch configuration and task delivery |
| Firmware Management | Supports batch firmware upgrades and task tracking |
| Remote Web Access | Opens supported router Web interfaces for remote configuration |
| Operations Visibility | Provides connection, carrier-traffic, task, location, and remote-traffic views |
| Client Access | Supports Windows, Linux, Android, and iOS clients with login and access controls |
| Peer-to-Peer Networking | Connects authorized users to remote devices through virtual-IP or real-IP access |
| Star Networking | Provides authorized remote access to designated enterprise resources |
| Network Permissions | Controls access by user, network group, site, and account role |
| Connection Types | Supports user-to-device, site-to-site, device-to-device, and multi-site communication |
| Deployment | Establishes secure connections without requiring public IP addresses or dedicated VPN firewalls |
| Access Capacity | Supports up to 4,000 router access points per account and 254 terminal devices per on-site router node |
| Connection Security | Uses access policies and encrypted transmission to protect remote connections |

## 5 Major R&D Centers

Rapid Response to Market Demands



Beijing



Toronto
Canada

Chengdu

Jiaxing

Virginia
USA







## 2 Major Manufacturing Centers

Supply Chain: Ensured Production Capacity and Quality



**2** **Manufacturing Centers**
Jiaxing (China)
Toronto (Canada)

**15** high-efficiency production lines

**11w+** monthly capacity

## 4 Major Warehousing and Technical Support Centers

Multi-location Warehousing and Technical Support

Toronto
Canada

Virginia
USA

Dusseldorf
Germany

Jiaxing
China



## Used worldwide. Proven worldwide.



**Canada**
Management of
Hydropower Station



**Germany**
Smart
EV Charging Kiosks



**China**
Predictive Maintenance
of Air Compressors



**US**
Predictive Maintenance
of Generators



**Switzerland**
Predictive Maintenance
of Textile Machines



**Thailand**
Flood Early Warning



**Australia**
Wireless Water Metering

**InHand** Networks

3650 Concorde Pkwy, Suite 200

Chantilly, VA 20151

USA

T: +1 (703) 348-2988

E: info@inhand.com

www.inhand.com



/ inhandnetworks

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